

Name: _____

Date: _____

SURDS, (MULTIPLYING AND DIVIDING SURDS)

LO: I can multiply and divide surds using the rules of surds.

Multiplying and dividing surds

To **multiply surds**: $a \times b = (ab)$. To **divide surds**: $a \div b = (a/b)$.

Key fact: $a \times a = a$. Simplify the result where possible.

Quick examples

●	$3 \times 5 = 15$	multiply under the root
●	$6 \times 6 = 6$	square root times itself = integer
●	$20 \div 5 = 4 = 2$	divide under the root
●	$3 \times 5 = 8$	multiply, not add
●	$2 \ 3 \times 4 \ 5 = 6 \ 15$	should be $8 \ 15$

Key idea

Multiply or divide the numbers under the root sign. Multiply coefficients separately.

$$\textcircled{a} \times \textcircled{b} = (a \times b)$$

Same rule works for division: $a \div b = (a \div b)$

MODELLED EXAMPLES

Study each example carefully before attempting the practice questions.

Example 1 - multiplying simple surds

Calculate 3×7 .

Multiply under the root: $3 \times 7 = 21$

$$3 \times 7 = 21$$

21 cannot be simplified

21

Multiply the radicands together.

Example 2 - with coefficients

Calculate $3 \ 2 \times 4 \ 5$.

Multiply coefficients: $3 \times 4 = 12$

Multiply surds: $2 \times 5 = 10$
 $= 12 \ 10$

12 10

Coefficients multiply separately from the surds.

Example 3 - simplifying the product

Calculate 6×8 .

$$6 \times 8 = 48$$

$$48 = (16 \times 3) = 4 \ 3$$

Simplified: $4 \ 3$

4 3

Always check if the product simplifies.

Example 4 - dividing surds

Calculate $72 \div 2$.

$$72 \div 2 = (72 \div 2) = 36$$

$$36 = 6$$

Answer is a whole number

6

When dividing, divide the numbers under the root.

YOUR TURN – PRACTICE (deliberate practice)

1. Multiplying simple surds

Multiply and simplify where possible.

- a) 2×3
- b) 5×5
- c) 7×3
- d) 2×8

2. With coefficients

Multiply and simplify.

- a) $2 \sqrt{3} \times 5 \sqrt{3}$
- b) $3 \sqrt{2} \times 2 \sqrt{7}$
- c) $4 \sqrt{5} \times 5$
- d) $2 \sqrt{6} \times 3 \sqrt{2}$

3. Dividing surds

Divide and simplify.

- a) $12 \div \sqrt{3}$
- b) $50 \div \sqrt{2}$
- c) $6 \sqrt{10} \div 2 \sqrt{5}$
- d) $75 \div \sqrt{3}$

4. Expanding brackets with surds

Expand and simplify.

- a) $3(\sqrt{3} + 2)$
- b) $5(2\sqrt{5} - 1)$
- c) $2\sqrt{2}(\sqrt{2} + \sqrt{3})$
- d) $7(\sqrt{3} - \sqrt{7})$

5. Mixed multiplication and division

Simplify fully.

- a) $3 \times \sqrt{12}$
- b) $(2\sqrt{3})^2$
- c) $18 \div \sqrt{6}$
- d) $(1 + \sqrt{2})(1 - \sqrt{2})$
- e) $(\sqrt{3} + \sqrt{5})^2$

COMMON MISCONCEPTIONS – SPOT THE MISTAKE

Each statement shows a student's answer. Tick () the ones that are correct. If it is wrong, write the correct answer.

a) $3 \times \sqrt{5} = 15$ ☐

Correct answer: _____

Reason: _____

b) $3 \times \sqrt{5} = 8$ ☐

Correct answer: _____

Reason: _____

c) $a \times a = a$ ☐

Correct answer: _____

Reason: _____

d) $2\sqrt{3} \times 4\sqrt{5} = 6\sqrt{15}$ ☐

Correct answer: _____

Reason: _____

e) $72 \div \sqrt{2} = 36 = 6$ ☐

Correct answer: _____

Reason: _____

6. CHALLENGE – WORD PROBLEMS (apply your skills)

a) Show that $(\sqrt{5} + \sqrt{3})(\sqrt{5} - \sqrt{3}) = 2$. Use this result to explain why the product of conjugate surds is always rational.

Expression: _____

Simplified: _____

b) A rectangle has area $12\sqrt{6} \text{ cm}^2$ and width $2\sqrt{3} \text{ cm}$. Find the length in simplified surd form. Find the perimeter of the rectangle.

Expression: _____

Simplified: _____

TOP TIPS

Read the question carefully. Show all your working step by step.
Check your answer makes sense.

CHECK YOUR WORK

Have I shown all my working clearly?

Did I check my answer using a different method?

Does my answer look reasonable?

